Hole Injection at Top and Edge Metal Contacts to Few-Layer InSe: A Verified Drift-Diffusion and Electro-Thermal Simulation Study

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Abstract

Efficient hole injection is the less-studied half of complementary logic in indium selenide (InSe). We simulate hole transport through 5-nm γ-InSe channels contacted by top (out-of-plane) and edge (in-plane) metal electrodes with an open-source DEVSIM model that couples Poisson's equation to the non-equilibrium carrier densities, uses a flux-balanced thermionic/thermionic-field-emission contact condition with a WKB van der Waals (vdW) gap transmission, and solves a lattice heat equation with quasi-Fermi-level Joule heating and explicit thermal boundary conditions. The implementation is verified against the Mott–Gurney law, thermionic-emission theory and energy conservation. We report total two-terminal resistance and decompose it exactly into contact and channel parts. Under Schottky–Mott alignment the hole barrier vanishes at $\Phi_M = E_V = 5.0$ eV by construction; with a pinning factor $S \leq 0.5$ no barrier collapse occurs below 5.4 eV, so the threshold is a property of the input alignment, not a prediction. For barrier-limited contacts ($\Phi_M = 4.8$ eV) the edge contact lowers total resistance by 58% relative to a 50-nm top contact, but almost all of this difference is due to the vdW tunnel gap: with the gap removed the two geometries differ by 4%. For barrier-free contacts the channel dominates ($\geq$86% of R_{total}), resistance scales as $L_{\text{ch}}^{2.0-2.3}$ because the channel is filled by contact-injected holes, and transfer-length extraction fails. Field-dependent tunnelling through the Schottky barrier changes resistance by at most 6% at $N_A = 10^{16}$ cm^{-3}. At high bias top contacts become injection-limited, so edge-contacted devices carry up to 5.4$\times$ more current and run hotter, both at equal bias and at equal dissipated power, unless heat is removed through the contacts.

Keywords: indium selenide; Schottky contact; hole injection; drift-diffusion; electro-thermal simulation; van der Waals gap

1 Introduction

InSe combines high electron mobility in thin, encapsulated flakes [1], near-ballistic short-channel transistors [2] and recently demonstrated wafer-scale growth [3], which makes it a candidate post-silicon channel [4]. Low-resistance electron contacts have been reported, for example with indium electrodes [5]; complementary logic, however, also requires hole injection into p-type InSe, which has received far less attention. Contacts to 2D semiconductors are generally controlled by three ingredients [6, 7]: the Schottky barrier set by metal work function and Fermi-level pinning; the tunnel barrier formed by the vdW gap between a top-deposited metal and the semiconductor; and the geometry through which current must spread from the contact into the channel. Edge contacts eliminate the vdW gap and inject in-plane, which is why they improve injection in graphene and other 2D systems [8, 9].

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This paper asks a narrow, simulation-answerable question: for a 5-nm p-type InSe channel, how do work function, pinning, vdW gap, contact geometry, channel length and self-heating each change the measured two-terminal resistance of top- and edge-contacted devices, and how much of that resistance actually sits at the contacts? An earlier version of this study was built on a model in which the carrier boundary condition, the Poisson coupling and the background doping were implemented incorrectly (Sec. 3.7); every result reported here was recomputed with a corrected and independently verified implementation. The main messages for contact design are: (i) the resistance threshold near 5.0 eV is an input of Schottky–Mott alignment and is removed by moderate pinning; (ii) the top-versus-edge difference in the barrier-limited regime is almost entirely the vdW tunnel gap, not current crowding; (iii) once the barrier vanishes, a 50-nm channel rather than the contact sets the resistance and its length scaling is strongly superlinear, so transfer-length methods are not applicable; and (iv) edge contacts trade lower resistance for a higher heat load per unit footprint.

2 Model

2.1 Geometry and equations

The device is a 5-nm InSe film (2D cross-section, results per unit width) with a channel of length $L_{\mathrm{ch}} = 50$ nm unless stated. Top contacts of length $L_c \in \{10, 20, 50\}$ nm lie on the upper surface; edge contacts cover the two vertical faces. The lower surface is the thermal boundary to the substrate; it is electrically insulating (no back gate). We solve, with DEVSIM, Poisson's equation with the full charge $\rho = q(p - n + N_D - N_A)$, Scharfetter–Gummel electron and hole continuity equations with Shockley–Read–Hall recombination, and (optionally) the lattice heat equation

$$\nabla \cdot (\kappa \nabla T) + H = 0, \qquad H = -\mathbf{J}_p \cdot \nabla \phi_p - \mathbf{J}_n \cdot \nabla \phi_n \ \ (\geq 0), \tag{1}$$

where $\phi_{p,n}$ are the quasi-Fermi potentials [10]. Mobility depends on temperature as $\mu(T) = \mu_{300}(T/300\,\mathrm{K})^{-1.5}$; carrier statistics are evaluated at 300 K (Boltzmann), which is the main simplification of the electro-thermal model. The background is p-type, $N_A = 10^{16}$ cm^{-3}.

2.2 Contacts

At each metal/InSe interface the electrostatic potential is fixed by the effective work function, $\psi_c = V + \chi - \Phi_{\mathrm{eff}} + E_g/2 + \frac{1}{2} V_t \ln(N_C/N_V)$, with $\Phi_{\mathrm{eff}} = \Phi_{\mathrm{CNL}} + S(\Phi_M - \Phi_{\mathrm{CNL}})$ ($S = 1$: Schottky–Mott; $\Phi_{\mathrm{CNL}} = 4.35$ eV, mid-gap, assumed). The hole barrier is $\phi_{Bp} = \chi + E_g - \Phi_{\mathrm{eff}}$. Carriers obey a flux (Robin) condition balanced against the drift-diffusion current arriving from the semiconductor,

$$J_{p,s} = q\, v_R\, \mathcal{T}_{\mathrm{vdW}}\, [1 + \Gamma(F)]\, (p_s - p_{\mathrm{eq}}), \qquad v_R = \frac{A^* T^2}{q N_V}, \quad p_{\mathrm{eq}} = N_V e^{-\phi_{Bp}/kT}, \tag{2}$$

and analogously for electrons, which is the thermionic-diffusion contact of Crowell and Sze [11]. $\mathcal{T}_{\mathrm{vdW}}$ is the WKB transmission of a rectangular vdW tunnel gap, $\exp(-2d_{\mathrm{vdW}}\sqrt{2m_0\Phi_{\mathrm{vdW}}}/\hbar)$, used only for top contacts ($d_{\mathrm{vdW}} = 0.30$ nm, $\Phi_{\mathrm{vdW}} = 2.0$ eV, $\mathcal{T}_{\mathrm{vdW}} = 0.013$; both assumed and varied in Sec. 3.3); edge contacts have $\mathcal{T}_{\mathrm{vdW}} = 1$. The thermionic-field-emission enhancement

$$\Gamma(F) = \frac{1}{kT} \int_0^{\phi_B} \exp\left[\frac{E}{kT} - \frac{4\sqrt{2m^*}E^{3/2}}{3q\hbar F} \right] dE \tag{3}$$

is computed from the local interface field F at every contact node (WKB through a triangular barrier, $m^* = 0.17\, m_0$) and updated self-consistently with the solution, in the spirit of local

tunnelling-contact models [12, 13]. For $\Phi_{\text{eff}} > E_V$ the Schottky–Mott barrier would be negative and the interface degenerate, which Boltzmann drift-diffusion cannot describe; such barriers are clipped to zero and flagged (shaded in Fig. 2).

2.3 Thermal boundary conditions

The substrate boundary is a Robin condition $-\kappa\partial_n T = h_{\text{sub}}(T - 300\,\text{K})$ with $h_{\text{sub}} = 1/(R_{\text{TBR}} + t_{\text{ox}}/\kappa_{\text{ox}}) = 1.35 \times 10^3$ W cm^{-2}K^{-1}, corresponding to a thermal boundary resistance $R_{\text{TBR}} = 10^{-8}$ m^{2}K/W in series with 90 nm SiO_2 ($\kappa_{\text{ox}} = 1.4$ W/mK) on an isothermal Si substrate. Metal contacts are adiabatic in the baseline; heat sinking through the contacts ($h_c = 10^4$ W cm^{-2}K^{-1}) is tested separately. Lateral heat spreading in the metal and substrate is not modelled beyond these lumped conductances. Interface (contact) dissipation, $J_s\,\Delta\phi$, is computed but not deposited in the lattice, so temperatures are slight underestimates; it is below 7% of IV at 2 V for all geometries (up to 16% at intermediate bias for the 10-nm top contact).

2.4 Parameters

Table 1 lists all inputs with their provenance. The values for E_g, χ and the effective masses are bulk or few-layer literature values applied to a 5-nm (about six-layer) film. Thickness dependence is expected mainly through E_g (which increases as the film thins [14]) and through dielectric screening and phonon transport; Sec. 3.6 gives the effect of each.

Table 1: Model parameters. M: measured (bulk or thick flakes); L: literature-derived (calculation/compilation); R: representative value within a literature range; A: assumed. The last column gives the expected effect of thickness-dependence or uncertainty.

Parameter	Value	Type	Effect of uncertainty
E_g	1.30 eV	M (bulk 1.26) → R	larger at 5 nm → E_V deeper, threshold up
χ	3.70 eV ($E_V = 5.0$ eV)	L [15]	shifts threshold 1:1
m_e, m_h (DOS)	0.07, 0.17 m_0	L [16]	N_V, v_R, A^*
m^*_{tun}	0.17 m_0	A ($= m_h$)	heavier → smaller Γ
ε_r	6.4	L	weak (Fig. 8)
μ_n, μ_p (300 K)	1000, 50 cm^2/Vs	M [1], A	$R_{\text{ch}} \propto 1/\mu_p$
$\mu(T)$ exponent	1.5	A	self-heating feedback
$\tau_{n,p}$	1 ns	A	negligible
N_A	10^{16} cm^{-3}	A	factor $\leq$4 on R over 10^{15}–10^{17}
κ_{InSe}	0.085 W/cm K	R	weak on $T_{\max}$
h_{sub}	1.35×10^3 W/cm^{2}K	A	dominant for $T_{\max}$
d_{vdW}, Φ_{vdW}	0.30 nm, 2.0 eV	A	dominant for top-contact R_c
S, Φ_{CNL}	1.0, 4.35 eV	A	dominant for barrier height
N_C, N_V, n_i	4.65×10^{17}, 1.76×10^{18}, 1.09×10^7 cm^{-3}	computed	–
v_R (holes)	6.5×10^6 cm/s	computed	–

2.5 Extracted quantities

$R_{\text{tot}} = V/I$ is the total two-terminal resistance (Ω·cm, per unit width) at the stated bias; the default is $V_{DS} = 0.05$ V, and the differential resistance dV/dI is reported where the I–V is nonlinear. R_{tot} is decomposed exactly using the hole quasi-Fermi potential on the film centre line: $R_{c,s} = \phi_p(x_0)/I$, $R_{c,d} = [V - \phi_p(x_1)]/I$ and $R_{\text{ch}} = [\phi_p(x_1) - \phi_p(x_0)]/I$, where x_0, x_1 are the inner contact edges (top) or the contact faces (edge), so $R_{\text{tot}} = R_{c,s} + R_{\text{ch}} + R_{c,d}$ identically. R_c therefore contains the interface jump plus the under-contact access path; the interface part alone is $R_{\text{if}} = P_{\text{if}}/I^2$. Temperature, resistance and power are always reported at the same operating point.

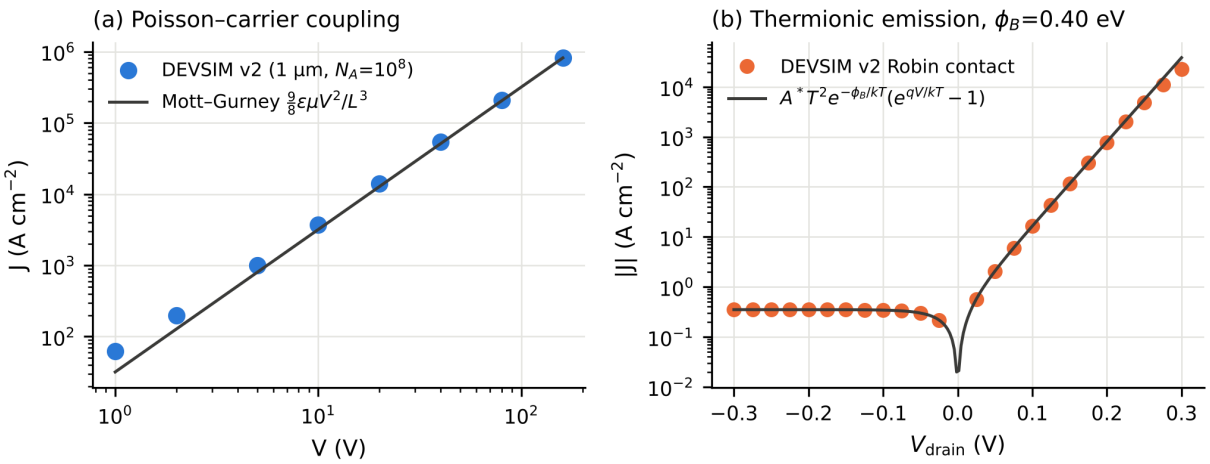


Figure 1: Verification. (a) Mott–Gurney space-charge-limited current in a 1-μm, nearly intrinsic film with barrier-free contacts. (b) Schottky diode with $\phi_B = 0.40$ eV and high mobility compared with thermionic-emission theory.

2.6 Verification and numerics

Before use, the implementation was checked against closed-form results (Fig. 1, data in `verification_summary.csv`): (V1) a uniformly doped p-type resistor reproduces $L/(qN_A\mu_p)$ to 2×10^{-14}; (V2) with finite interface velocity v it reproduces $L/(qN_A\mu_p) + 2V_t/(qvN_A)$ within 1.2×10^{-3}, confirming that Eq. (2) acts as a flux condition; (V3) an undoped 1-μm film with barrier-free contacts approaches the Mott–Gurney law, $J/J_{\mathrm{MG}} = 1.05$, 1.03 and 1.008 at 40, 80 and 160 V with $d\ln J/d\ln V = 1.97$, confirming the Poisson–carrier coupling; (V4) a 0.40-eV Schottky diode gives ideality 0.994 and reverse saturation within 0.12% of $A^*T^2e^{-\phi_B/kT}$; (V5) in 2D electro-thermal runs $IV = P_{\mathrm{bulk}} + P_{\mathrm{if}}$ to 10^{-12} and the heat leaving the boundaries equals P_{bulk} within 0.5%. R_{tot} changes by less than 1.3% between 0.5× and 3× mesh density (910–27 820 nodes for the top contact, 279–7 697 for the edge contact; Fig. S1), so all results use the 1× mesh. The node-level Γ at the contact corner is mesh dependent (a field singularity), but its effect on R_{tot} is within the same 1.3%.

3 Results

3.1 Work function, pinning and the barrier threshold

Figure 2(a) shows the hole barrier used as input. With $S = 1$ it falls linearly and reaches zero at $\Phi_M = \chi + E_g = 5.0$ eV. R_{tot} [Fig. 2(b)] follows it exponentially, from 1.1×10^{10} (edge) and 4.9×10^{10} $\Omega \cdot$ cm (top, 50 nm) at 4.33 eV to 11.3 and 15.6 $\Omega \cdot$ cm at 5.0 eV; a fine sweep (4.90–5.00 eV in 25-meV steps) shows the approach is smooth, with R_{tot} falling by a factor of 4.4 (edge) and 6.2 (top 50 nm) over the last 0.1 eV. Above 5.0 eV the curves are flat only because the barrier is clipped. This threshold is therefore a direct consequence of the adopted Schottky–Mott alignment and the 3.7-eV electron affinity, not an independent prediction. With pinning it moves or disappears: for $S = 0.5$ the barrier at 5.0 eV is 0.33 eV and $R_{\mathrm{tot}}(\mathrm{edge}) = 1.8 \times 10^5$ $\Omega \cdot$ cm, and for $S = 0.3$ no metal in the 4.3–5.4 eV range gives a barrier below 0.33 eV (`pinning_sweep.csv`). Interface dipoles, metal-induced gap states and thickness-dependent χ and E_g enter in the same

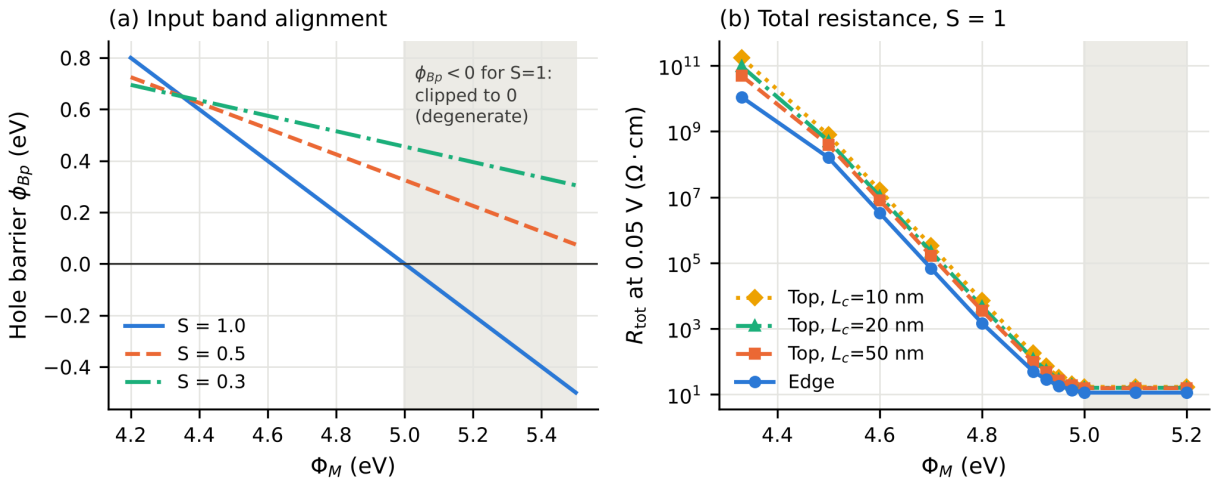


Figure 2: (a) Hole barrier versus metal work function for pinning factors $S = 1$, 0.5 and 0.3. (b) R_{tot} at 0.05 V for top ($L_c = 10$, 20, 50 nm) and edge contacts, $S = 1$. In the shaded region the Schottky–Mott barrier is negative and has been clipped to zero.

way, through Φ_{eff} and E_V: a 0.1-eV shift of E_V changes R_{tot} at 4.8 eV by a factor of about 25–50 (Fig. 8). We note that the near-ohmic indium electron contacts of Ref. [5] would correspond, in the same Schottky–Mott picture with $\chi = 3.7$ eV, to a 0.6-eV electron barrier, which suggests that interface chemistry or pinning plays a significant role in real InSe contacts; we therefore make no quantitative comparison with that experiment, which also concerns electron rather than hole transport.

3.2 Where the resistance is: contact versus channel

Figure 3 decomposes R_{tot}. In the barrier-limited case ($\Phi_M = 4.8$ eV, $\phi_{Bp} = 0.2$ eV) the contacts carry 55–76% of R_{tot} for top contacts, mostly at the interface itself, and only 7% for the edge contact. The small edge value does not mean the barrier is unimportant: the 50-nm channel is much shorter than the depletion width at $N_A = 10^{16}$ cm^{-3} (about 0.1 μm for 0.2 eV), so the band bending, and hence the voltage drop, extends across the whole channel and is booked as "channel". Shorter top contacts are worse (7130, 4680 and 3530 $\Omega \cdot$cm for $L_c = 10$, 20 and 50 nm), as expected when the interface conductance per unit length is low. In the barrier-free case ($\Phi_M = 5.0$ eV) the channel carries 86–93% (top) and 99.5% (edge) of R_{tot}; $2R_c = 1.17\ \Omega \cdot$ cm (top 50 nm) and 0.056 $\Omega \cdot$cm (edge). These decomposed contact resistances are independent of L_{ch} across 25–200 nm to within 1% at 5.0 eV and within 6% (top) and 12% (edge) at 4.8 eV (`lch_sweep.csv`), which is the expected behaviour of a contact quantity and an internal check of the decomposition. By contrast, a linear transfer-length fit of $R_{\mathrm{tot}}(L_{\mathrm{ch}})$ gives negative intercepts ($-60\ \Omega \cdot$ cm) at 5.0 eV and, at 4.8 eV, intercepts 1.6 (top) to 10 (edge) times larger than $2R_c$, because R_{tot} is not linear in L_{ch} in either regime [Fig. 5(c)]. Transfer-length extraction should therefore not be applied to such devices.

3.3 Top versus edge: the role of the vdW gap and of tunnelling

At 0.05 V the edge contact has lower R_{tot} than every top contact at every work function: by 58% (4.8 eV) and 27% (5.0 eV) relative to the 50-nm top contact. Figure 4(c) shows where this comes

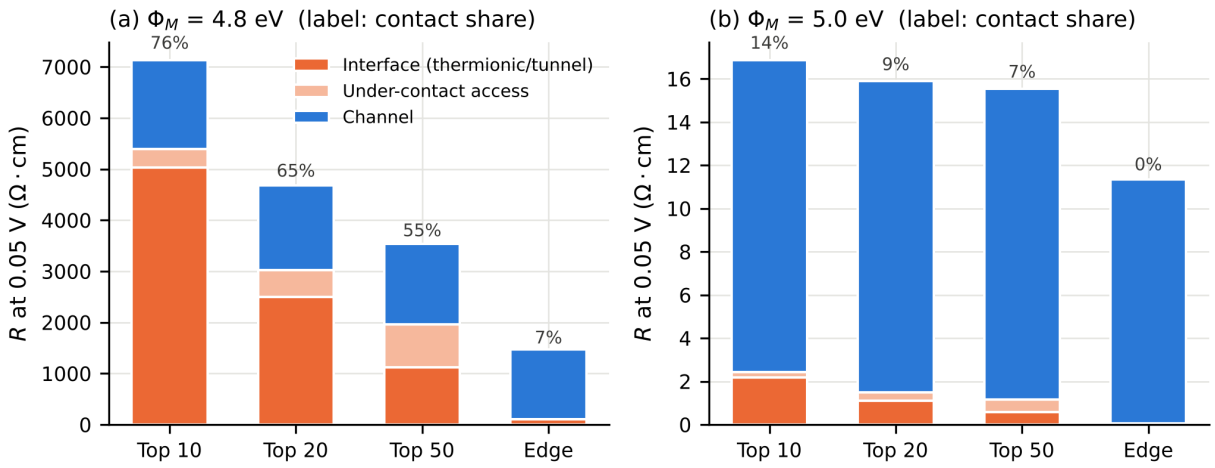


Figure 3: Exact decomposition of R_{tot} at 0.05 V into interface, under-contact access and channel contributions at (a) $\Phi_M = 4.8$ eV and (b) 5.0 eV. Labels: contact share $2R_c/R_{\mathrm{tot}}$.

from. Removing the vdW gap ($\mathcal{T}_{\mathrm{vdW}} = 1$) lowers the 50-nm top-contact resistance at 4.8 eV from 3530 to 1540 $\Omega \cdot$ cm, within 4% of the edge value (1480 $\Omega \cdot$ cm); widening it to 0.5 nm raises it to 2.4×10^4 $\Omega \cdot$cm. At 5.0 eV the gap-free top contact is still 22% more resistive than the edge contact (14.6 vs 11.3 $\Omega \cdot$ cm), a residual geometric effect of the longer access path and smaller injected hole density in the channel. In the barrier-limited regime, then, the top/edge difference is essentially the vdW tunnel barrier, and its magnitude is only as reliable as the assumed d_{vdW} and Φ_{vdW}.

Tunnelling through the Schottky barrier itself is small at this doping. The field-dependent $\Gamma(F)$ lowers R_{tot} by at most 6.1% (edge, 4.33 eV) and by 2–4% at 4.6–4.8 eV [Fig. 4(a)]; at $N_A = 10^{16}$ cm^{-3} the interface field is $\sim 10^4$–10^5 V/cm and $\Gamma \sim 0.2$–2. Because Γ now enters a flux condition, it does change the solution when it is large: imposing a constant $\Gamma = 100$ (the kind of value the earlier version tested) lowers the 50-nm top-contact resistance at 4.8 eV by 47% and the edge resistance by 6% [Fig. 4(b)]. The interface velocity therefore matters most where the interface conductance is small, that is, for top contacts with a vdW gap. A nonlocal tunnelling model and heavier doping near the contacts would be needed for quantitative statements about field emission.

3.4 Channel transport in the barrier-free regime

With barrier-free contacts $p_{\mathrm{eq}} = N_V = 1.76 \times 10^{18}$ cm^{-3}, and holes spill from both contacts into the channel. Even at 0.05 V the centre-line hole density of a 100-nm edge-contacted channel exceeds N_A by a factor of 2.3 at mid-channel and by more than 100 within a few nanometres of the contacts [Fig. 5(b)]. Consequently R_{tot} scales as $L_{\mathrm{ch}}^{2.3}$ (edge) and $L_{\mathrm{ch}}^{2.0}$ (top, 50 nm) over 25–200 nm at 0.05 V [Fig. 5(c)], close to the L^{-2} conductance scaling of space-charge-limited current in thin films [17], whereas at 4.8 eV the length dependence is weak ($L^{0.1}$–$L^{0.3}$) because the barrier, not the channel, controls the current. The bias dependence, however, does not reach a developed space-charge-limited regime. The local exponent $m = d\ln I/d\ln V$ of isothermal edge-contacted devices rises from 1.0 at low bias to a maximum of 1.5–1.7 [Fig. 5(a)]; power-law fits between 0.4 and 1.2 V give $m = 1.47 \pm 0.02$, 1.44 ± 0.02 and 1.28 ± 0.02 for $L_{\mathrm{ch}} = 50$, 100 and 200 nm (one standard error of the fit). The 25-nm device instead saturates ($m < 1$) as its current approaches the thermionic injection limit $qv_R N_V t = 0.92$ A/cm. We therefore describe the barrier-free channel as contact-injection-dominated, with superlinear I–V and SCLC-like length scaling, but not as a demonstrated

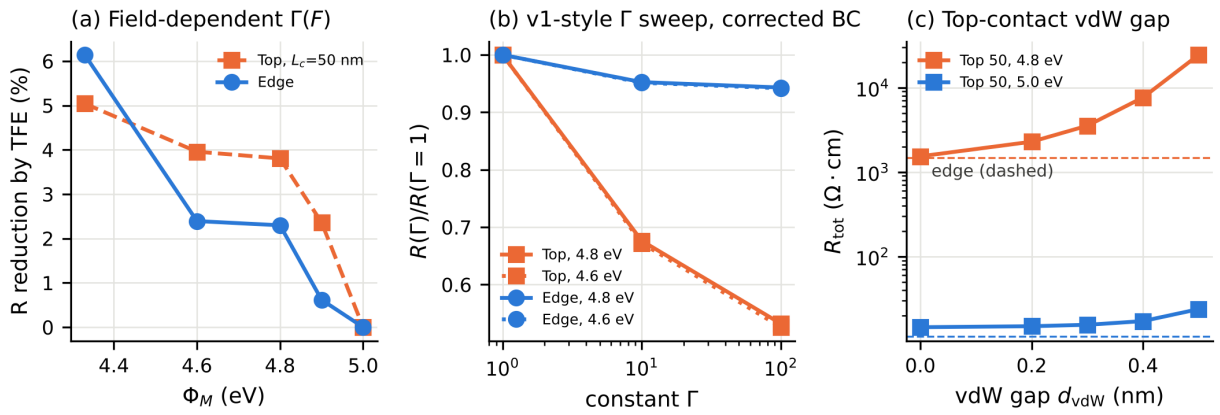


Figure 4: (a) Reduction of R_{tot} by the self-consistent, field-dependent TFE factor $\Gamma(F)$. (b) R_{tot} for an imposed constant Γ, normalized to $\Gamma = 1$, in the corrected flux boundary condition. (c) 50-nm top-contact R_{tot} versus vdW-gap width; dashed lines: edge contact.

Mott–Gurney or thin-film SCLC regime: the exponents are below 2, the background doping is not negligible, and diffusion from both contacts contributes at all biases. Profiles of p, ψ, E and ϕ_p for all cases are provided in `profiles.npz`.

3.5 High bias and self-heating

Figure 6 and Table 2 compare the geometries at identical operating points. At 5.0 eV the I–V of top contacts saturates: the current approaches the vdW-limited injection capacity $qv_R\mathcal{T}_{\mathrm{vdW}}N_V L_c$ (0.024 A/cm for $L_c = 10$ nm, approached by 0.5 V; 0.12 A/cm for 50 nm, of which current crowding uses about half), and dV/dI rises to 10^2–$10^3\ \Omega\cdot\mathrm{cm}$. Edge contacts do not saturate and carry 0.35 A/cm at 2 V, 5.4× the 50-nm top contact. The resulting temperature rise is larger for the edge contact both at equal bias (105 K vs 7.8 K at 2 V) and at equal dissipated power: the peak rise per unit power is 152 K cm/W for the edge contact against 61 (top, 50 nm) and 102 K cm/W (top, 10 nm), because the edge-contacted film has a 50-nm thermal footprint on the substrate whereas the top-contacted film also spreads heat under its contacts. Within the model, therefore, edge contacts do not run cooler; their advantage is lower resistance and higher current, and their thermal penalty depends on how heat leaves the device. The peak temperature is set mainly by the substrate heat-transfer coefficient (edge, 2 V: 441 K rise for h_{sub} ten times smaller, 16 K for ten times larger) and by whether metal contacts sink heat (55 K with $h_c = 10^4$ W cm^{-2}K^{-1}), and only weakly by κ_{InSe} (103–115 K for 0.02–0.2 W/cm K); absolute temperatures are therefore model-dependent until h_{sub} and h_c are calibrated for a specific stack. At 4.8 eV self-heating is below 0.3 K up to 2 V for all geometries.

3.6 Parameter sensitivity

Figure 8 summarizes the change of R_{tot} at 0.05 V when one parameter is varied. In the barrier-limited case the band alignment dominates: ±0.1 eV in E_V changes R_{tot} by factors of 0.04–47, and raising E_g to 1.45 eV at fixed χ (a plausible value for thinner films) raises it by more than two orders of magnitude. μ_p, m_h and N_A change R_{tot} by factors of about 2–4, and ε_r by less than 2%. In the barrier-free case R_{tot} is roughly proportional to $1/\mu_p$, and a 0.15-eV deeper E_V (via

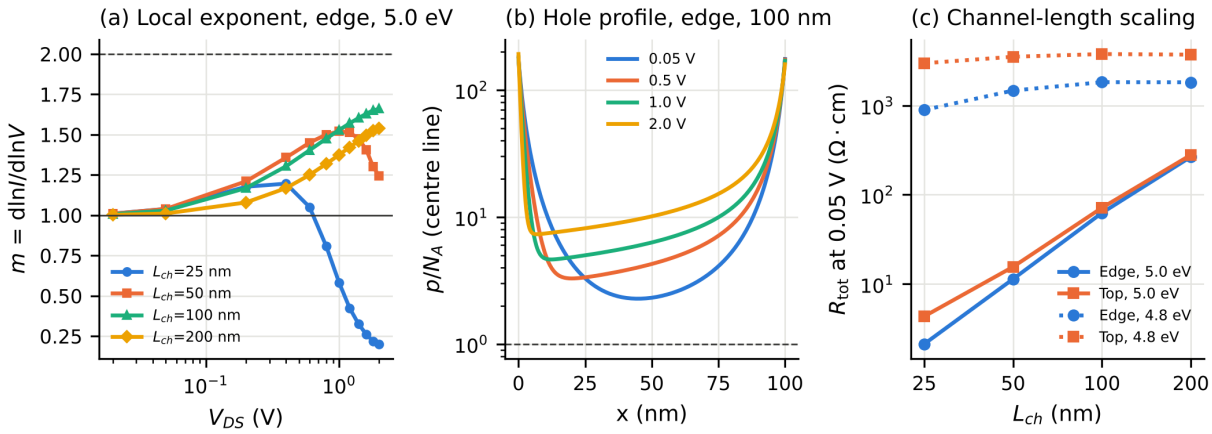


Figure 5: Barrier-free ($\Phi_M = 5.0$ eV) channel transport. (a) Local I–V exponent for edge-contacted channels of 25–200 nm (isothermal). (b) Centre-line hole density of a 100-nm edge-contacted channel at 0.05–2 V. (c) R_{tot} at 0.05 V versus channel length at 4.8 and 5.0 eV.

Table 2: Operating points at $\Phi_M = 5.0$ eV, $L_{\mathrm{ch}} = 50$ nm, baseline thermal model. All quantities in a row refer to the same bias.

Geometry	V_{DS} (V)	I (A/cm)	R_{tot} ($\Omega \cdot$ cm)	dV/dI ($\Omega \cdot$ cm)	IV (W/cm)	ΔT_{max} (K)
Top, 10 nm	0.05	0.00297	16.9	16.7	1.5×10^{-4}	0.0
Top, 50 nm	0.05	0.00322	15.6	15.3	1.6×10^{-4}	0.0
Edge	0.05	0.00441	11.3	11.1	2.2×10^{-4}	0.0
Top, 10 nm	2.0	0.0233	85.8	4700	0.047	4.8
Top, 50 nm	2.0	0.0641	31.2	262	0.128	7.8
Edge	2.0	0.346	5.78	6.77	0.692	105.4

$E_g = 1.45$ eV) reintroduces a barrier and raises R_{tot} by a factor of 20–35. Background doping is now a moderate parameter (factor $\leq$4 between 10^{15} and 10^{17} cm^{-3}).

3.7 Relation to the earlier version of this work

The earlier version (repository folder `src/`) differed in five implementation points, all corrected here: the contact expression $qv(1+\Gamma)(p-p_{\mathrm{eq}})$ was used as the residual of a Dirichlet-type contact equation, which enforces $p_s = p_{\mathrm{eq}}$ for any nonzero prefactor, so Γ and $\mathcal{T}_{\mathrm{vdW}}$ could not affect the result; Poisson's equation retained the equilibrium charge after drift-diffusion was enabled; the background was n-type ($+10^{16}$ cm^{-3}) instead of p-type; the contact potential carried a sign error in the $\ln(N_C/N_V)$ term; and resistance at 0.1 V was paired with temperature at 2 V. The earlier conclusions that tunnelling is irrelevant because resistance was invariant to Γ, that the barrier-free channel is SCLC, that the edge advantage is geometric, and that edge contacts run 94 K cooler, do not hold in the corrected model and are withdrawn.

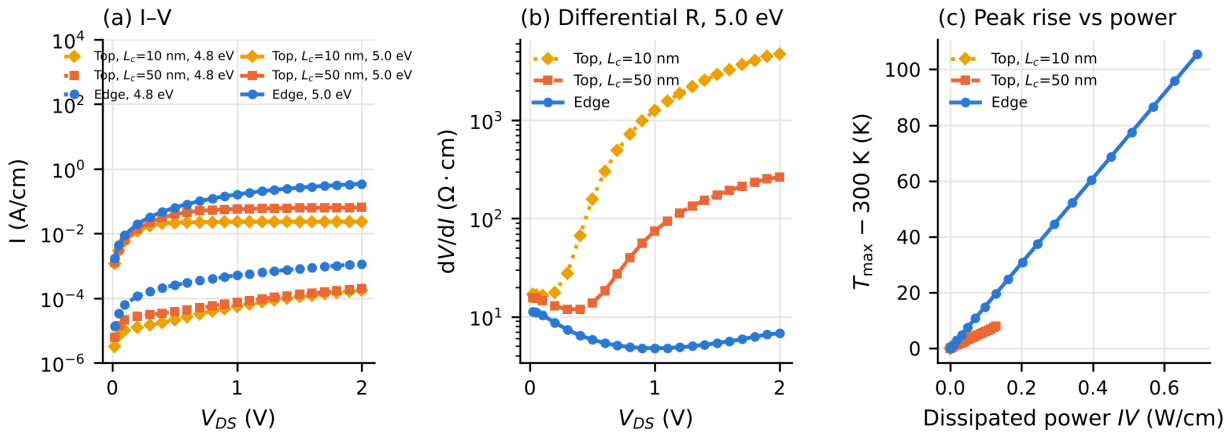


Figure 6: Electro-thermal I–V. (a) Current versus bias at 4.8 (dotted) and 5.0 eV (solid). (b) Differential resistance at 5.0 eV. (c) Peak temperature rise versus dissipated power IV at 5.0 eV.

4 Conclusions

For hole injection into a 5-nm InSe channel, the corrected and verified model gives four results. First, the resistance threshold near 5.0 eV is the Schottky–Mott valence-band edge of the input alignment and disappears for pinning factors $S \leq 0.5$; contact-metal selection for p-type InSe hinges on the real barrier height, which must come from experiment. Second, for barrier-limited contacts the top-versus-edge difference (58% at 4.8 eV) is almost entirely the vdW tunnel gap; without it the geometries differ by 4%. Third, for barrier-free contacts a 50-nm channel, filled by contact-injected holes, carries 86–99.5% of the resistance and scales superlinearly with length, so contact resistance must be extracted by decomposition rather than by transfer-length fits. Fourth, top contacts become injection-limited at high bias, while edge contacts deliver up to 5.4× more current and a larger temperature rise per unit power, so edge-contacted InSe benefits most from heat removal through the contacts and substrate. Quantitative predictions for specific metals require measured barrier heights, vdW-gap parameters and thermal boundary conductances.

Statements and Declarations

Acknowledgements Simulations used DEVSIM, an open-source TCAD tool. I thank the anonymous reviewers of an earlier version, whose code-level reading identified the implementation errors corrected here.

Author contributions H.A. designed the study, developed the simulation code, performed all simulations and analyses, and wrote the manuscript.

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Competing interests The author declares no competing interests.

Data and code availability Code (`src_v2/`: model, verification, studies, figures), all data (`data_v2/`) and figures are available at https://github.com/Brukrish2006/

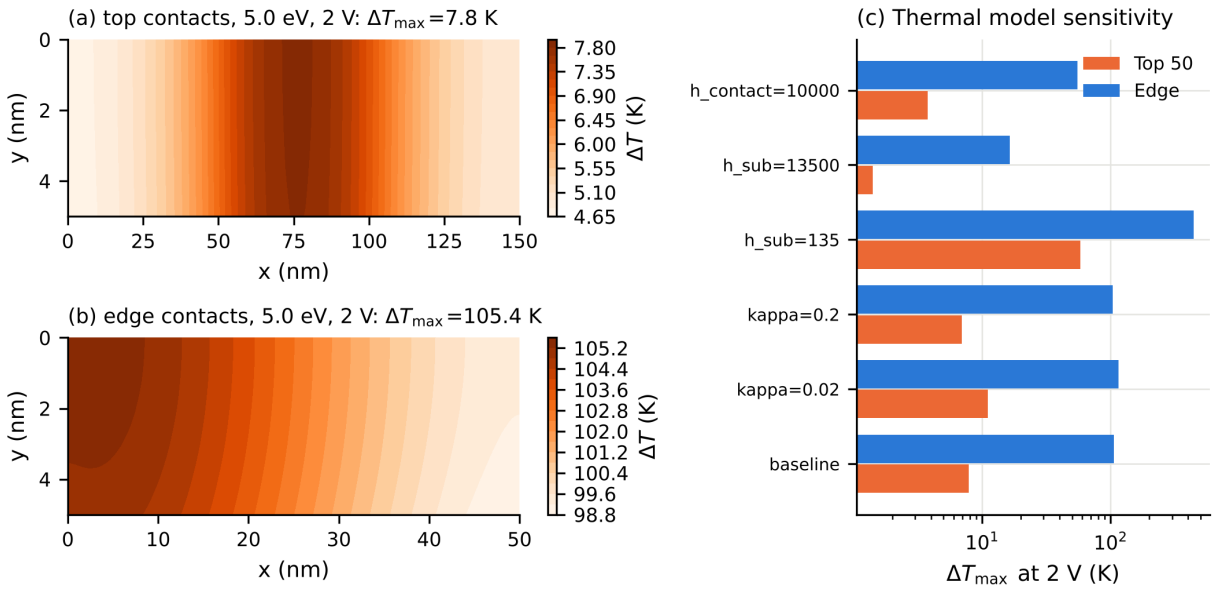


Figure 7: Lattice temperature rise at 2 V, 5.0 eV for (a) top ($L_c = 50$ nm) and (b) edge contacts, and (c) peak rise for variations of κ_{InSe} (W/cm K), h_{sub} and contact heat sinking h_c (W cm^{-2}K^{-1}).

InSe-Top-Edge-Contact-Resistace. The earlier implementation is retained in `src/` and `data/` for transparency.

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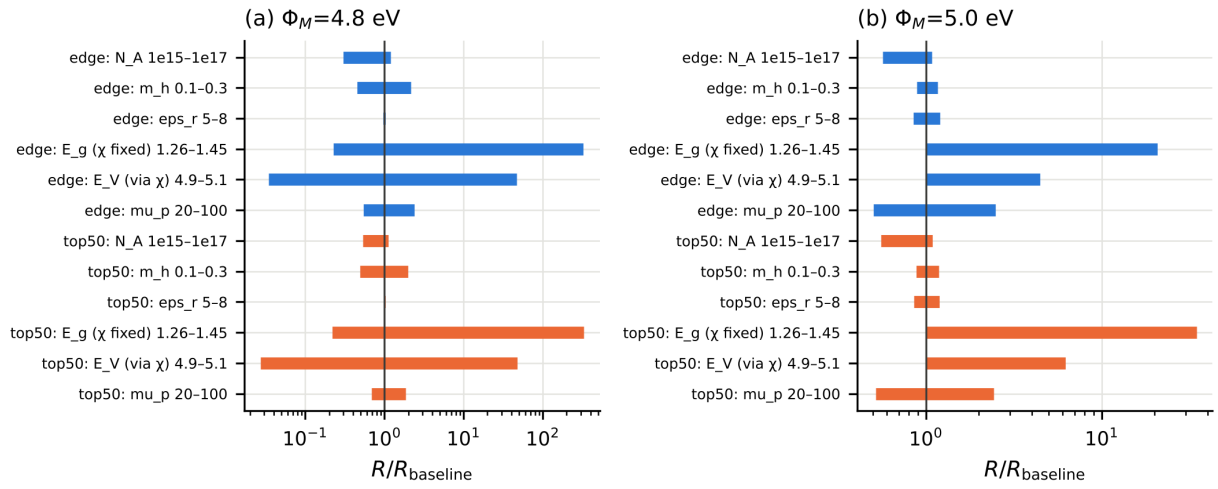


Figure 8: Range of $R_{\mathrm{tot}}/R_{\mathrm{baseline}}$ at 0.05 V for one-at-a-time parameter variations at (a) 4.8 and (b) 5.0 eV.

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Supplementary figure

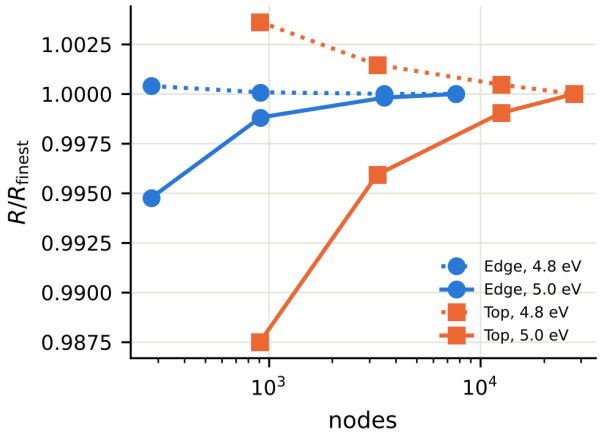


Figure S1: Mesh convergence of R_{tot} at 0.05 V (0.5×–3× density), normalized to the finest mesh.